DAE- Final Term Report

Project: "The Impact of Promotions, Inflation, and Regional Conditions on Weekly Retail Sales"

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1. Introduction

Retail businesses operate in a complex environment where sales are influenced by multiple factors, including promotions, macroeconomic conditions, regional characteristics, and seasonal trends. Understanding these influences and their interactions is essential for making informed decisions on inventory management, pricing strategies, promotional planning, and overall business performance.

This study, titled ***"The Impact of Promotions, Inflation, and Regional Conditions on Weekly Retail Sales"***, analyzes historical weekly sales data from 45 stores across multiple departments to uncover patterns, seasonal effects, and key predictors of sales. The dataset covers the period from February 2019 to November 2021 and includes information on store characteristics, department types, macroeconomic indicators such as inflation (CPI), unemployment, and fuel prices, as well as promotions, holidays, and temporal features like year, month, and week number.

The primary objective of this study is to measure the effects of promotions, macroeconomic conditions, and regional differences on weekly sales while identifying temporal and seasonal trends. The analysis aims to provide actionable insights that can guide decisions on promotion timing, resource allocation, and sales forecasting.

2. Data Description

The dataset used in this study was obtained from **data.world** and was originally compiled by **Nadeesha Dilhani Hettikankanamage, Niusha Shafi Abady, Fiona Chatteur, Robert M.X. Wu, Jianlong Zhou,** and **James Vakilian.** It contains historical sales data for 45 Amazon stores across different regions from February 5, 2019, to November 1, 2021. Each store is divided into multiple departments, with the aim of predicting department-level weekly sales.

The data also records promotional markdown events, concentrated around major holidays such as Super Bowl, Labor Day, Thanksgiving, and Christmas. Holiday weeks are given extra weight to reflect their impact on sales.

The dataset includes anonymized store and department information, including store type and size, as well as the following key variables: date, store and department identifiers, weekly sales, store type, store size, temperature, fuel price, CPI, unemployment rate, holiday indicators, promotional markdowns, and time features such as year, month, and week.

The panel structure of the dataset, with multiple stores and departments observed over time, allows for both temporal analysis of trends and seasonal patterns, as well as cross-sectional comparisons across stores and departments.

3. Data Cleaning

The initial dataset contained 374,247 rows and 21 columns, including weekly sales, store and department identifiers, store characteristics, macroeconomic indicators, and promotional markdown information. Cleaning and standardizing the data was necessary to ensure accuracy and usability.

3.1 Data Type Conversions and Column Renaming

Some columns were converted to appropriate data types for analysis. The *IsHoliday* column, initially represented as 0 or 1, was converted to a Boolean type and labeled "Yes" or "No". Categorical identifiers such as store, department, and type were converted to categorical variables. The column *Total_MarkDown* was renamed *Promotional_Discount* to better reflect its meaning.

3.2 Removal of Redundant Columns

Columns that were duplicates or did not provide meaningful information were removed. For example, *Unnamed: 3* duplicated the date column and was dropped, along with other statistical summary columns like min, max, mean, median, and standard deviation.

3.3 Handling Missing and Zero Values

Although the dataset contained no explicit missing values, zeros were present in *Weekly_Sales* and *Promotional_Discount*. For *Promotional_Discount*, zeros before November 2020 represented missing markdown data rather than actual zero discounts. These were replaced with the average promotional discount for the corresponding store from the period after November 2020. For *Weekly_Sales*, zero values were replaced with the average sales for the corresponding store, department, month, and type based on non-zero observations. After these adjustments, both columns contained no remaining zeros.

3.4 Outlier Detection and Treatment

Outliers in key numerical columns were addressed using the interquartile range method. Values below $Q1 - 1.5IQR$ or above $Q3 + 1.5IQR$ were replaced with the median of the respective column. This applied to *Weekly_Sales*, *Promotional_Discount*, *Temperature*, and *Fuel_Price*. After this step, the dataset contained 364,583 rows and 27 columns, preserving most data while reducing extreme variation.

3.5 Consistency Checks

Dates were confirmed to be in proper datetime format. Categorical values such as store, department, type, and holiday indicators were consistent and correctly encoded. Duplicate rows and unnecessary columns were removed. After these checks, all columns had valid values, ensuring the dataset was ready for feature engineering and exploratory analysis.

4. Feature Engineering

To better understand sales patterns and support predictive modeling, several new features were created from the raw dataset. Feature engineering focused on categorizing continuous variables and encoding holiday weeks to reflect the seasonal and panel nature of the data.

4.1 Promotional Discount Binning

The Promotional_Discount column was categorized to group weeks based on the size of discounts. Initially, all observations were assigned a default category, No Promotion. For weeks with positive discounts, the 25th, 50th, and 75th percentiles were calculated, and four additional bins were created:

- **Low Promotion:** Discount ≤ 25th percentile
- **Medium Promotion:** 25th < Discount ≤ 50th percentile
- **High Promotion:** 50th < Discount ≤ 75th percentile
- **Very High Promotion:** Discount > 75th percentile

This new feature, Promotion_Bin, was converted to a categorical variable. It enables the analysis of sales responses to different levels of promotional intensity.

4.2 CPI / Inflation Binning

To analyze the effect of inflation on sales, the Consumer Price Index (CPI) was grouped into three categories. Using the 33rd and 66th percentiles:

- **Low Inflation:** CPI ≤ 33rd percentile
- **Moderate Inflation:** 33rd < CPI ≤ 66th percentile
- **High Inflation:** CPI > 66th percentile

This created the categorical feature Inflation_Bin, which helps examine how sales vary under different inflation conditions.

4.3 Unemployment Binning

The Unemployment column was binned similarly to CPI to explore the effect of labor market conditions on sales:

- **Low Unemployment:** ≤ 33rd percentile
- **Moderate Unemployment:** 33rd < Unemployment ≤ 66th percentile
- **High Unemployment:** > 66th percentile

The resulting Unemployment_Bin feature allows for comparison of sales patterns across different unemployment levels.

4.4 Fuel Price Binning

To study regional cost conditions, fuel prices were categorized based on the 33rd and 66th percentiles:

- **Low Fuel Price:** ≤ 33rd percentile
- **Medium Fuel Price:** 33rd < Fuel Price ≤ 66th percentile
- **High Fuel Price:** > 66th percentile

The FuelPrice_Bin feature captures how changes in fuel costs can influence retail sales, especially in regions with varying transportation expenses.

4.5 Store Size Binning

Store size was binned to account for differences in operational scale. Using the 33rd and 66th percentiles:

- **Small Store:** ≤ 33rd percentile
- **Medium Store:** 33rd < Size ≤ 66th percentile
- **Large Store:** > 66th percentile

The StoreSize_Bin feature highlights structural differences between stores and how these differences may affect sales and the impact of promotions.

4.6 Holiday Type Encoding

To measure the effect of major US holidays on sales, the Holiday_Type feature was created based on key dates:

- **Super Bowl:** 12-Feb-19, 11-Feb-20, 10-Feb-21, 8-Feb-18
- **Labor Day:** 10-Sep-19, 9-Sep-20, 7-Sep-21, 6-Sep-18
- **Thanksgiving:** 26-Nov-19, 25-Nov-20, 23-Nov-21, 29-Nov-18
- **Christmas:** 31-Dec-19, 30-Dec-20, 28-Dec-21, 27-Dec-18

All weeks were initially labeled Non-Holiday, and specific holiday weeks were then assigned the appropriate holiday type. The Holiday_Type column was converted into a categorical variable. This feature allows for analysis of holiday-driven sales spikes and their interaction with promotional campaigns.

5. Data Preparation for Exploratory Analysis

After creating the binned and encoded features, the dataset was divided into numerical and categorical columns. This separation allows for appropriate visualization techniques and provides clearer insights into the distributions and patterns of each variable type.

5.1 Numerical Columns

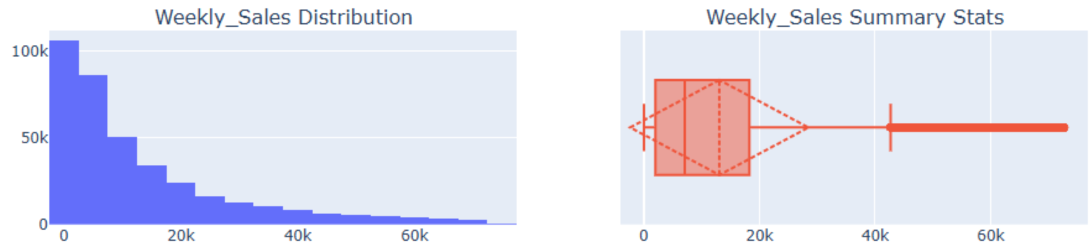
The following columns were treated as numerical:

- Weekly_Sales
- Size

- Temperature
- Fuel_Price
- CPI
- Unemployment
- Promotional_Discount

For these variables, histograms and boxplots were generated to examine their distributions, assess skewness and central tendency, and detect potential outliers.

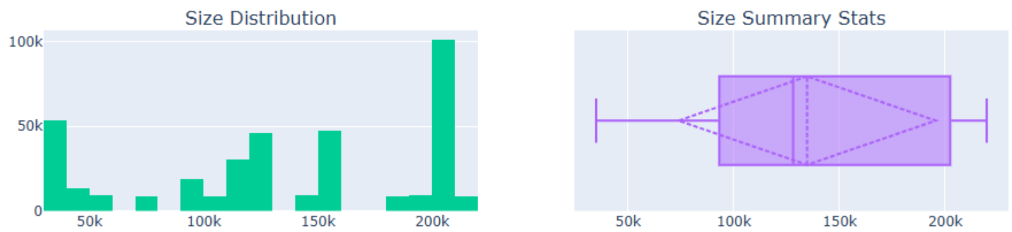
Weekly Sales:



Histogram: Weekly sales exhibit a highly right-skewed distribution, with most values concentrated between 0 and 20k and a long tail extending toward 80k.

Boxplot: The median is far to the left, confirming the skew. The long whisker to the right highlights high-performing weeks, likely driven by holidays or major promotions.

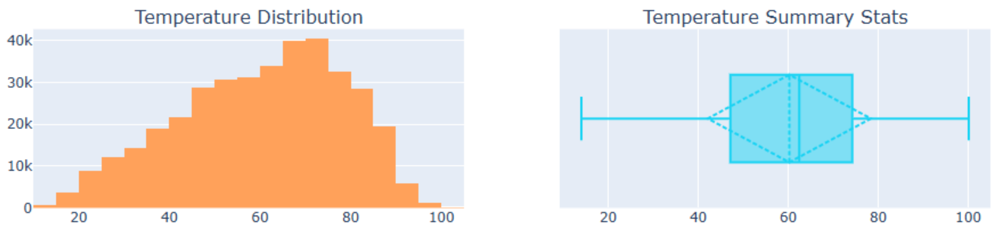
Store Size (Size):



Histogram: The distribution is tri-modal, with three clusters corresponding to small (~40k), medium (~125k), and very large stores (~200k+).

Boxplot: The wide interquartile range reflects the diversity in store formats. The mean is less representative than the peaks observed in the histogram.

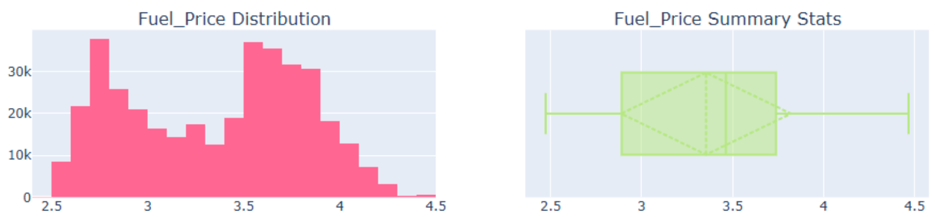
Temperature:



Histogram: Temperature follows an approximately normal distribution, slightly left-skewed, with most values between 40°F and 80°F.

Boxplot: The median is around 60°F, with balanced whiskers and few outliers, suggesting stores are in regions with moderate seasonal climates.

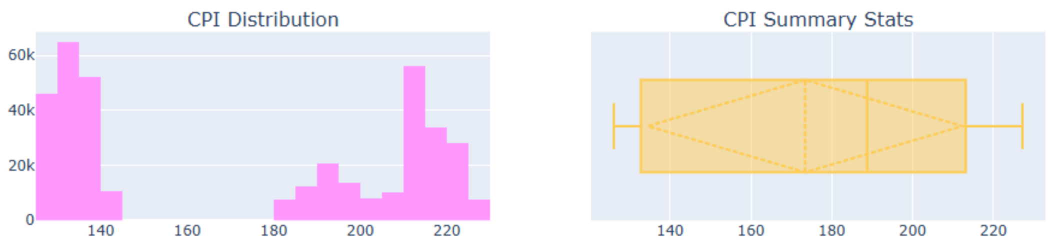
Fuel Price:



Histogram: The distribution is bimodal, with peaks around 2.7 and 3.7, indicating periods of significant shifts in fuel prices.

Boxplot: The median is about 3.4, while the bimodal nature shows the average falls in a less populated range.

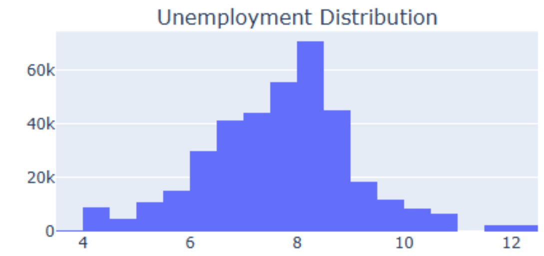
CPI (Consumer Price Index):

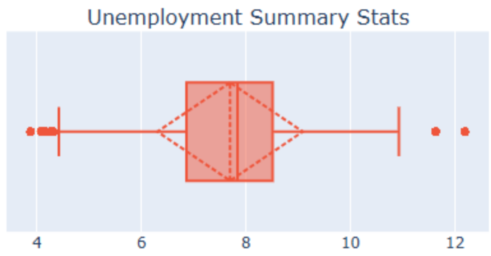


Histogram: CPI shows a highly bimodal distribution, with clusters at 130–140 and 210–225 and almost no values in 150–180. This likely reflects different regions or time periods.

Boxplot: The IQR spans a gap where few data points exist, making the median less informative.

Unemployment:

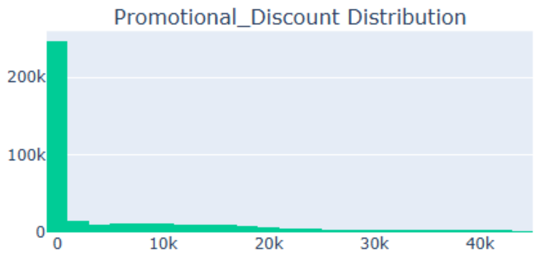


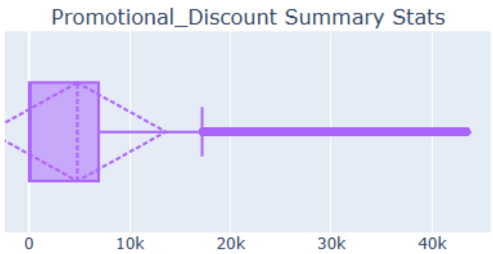


Histogram: Unemployment is unimodal, centred around 8%, with slight right skew and a small peak near 12%.

Boxplot: Most values fall between 7% and 9%, with outliers below 5% and above 12%, representing extreme economic conditions.

Promotional Discount:





Histogram: This is an extreme power-law distribution. Most weeks have no promotions, with rare instances of high discounts.

Boxplot: The median is near zero, while the long right tail shows occasional high-value promotions.

Key Takeaway:
Weekly sales and promotional discounts are driven by rare high-value events, while variables like CPI and store size indicate the dataset contains distinct categories or regions rather than a single uniform group.

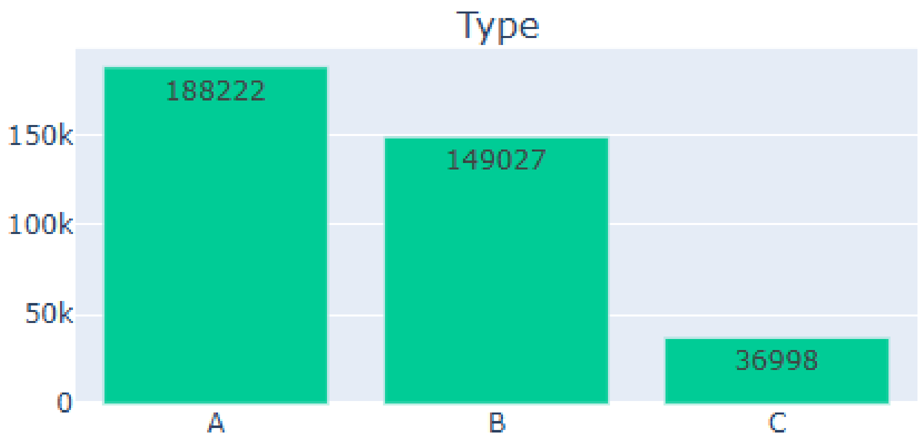
5.2 Categorical Columns

The following columns were treated as categorical:

- Type
- IsHoliday
- Year
- Month
- Promotion_Bin
- Inflation_Bin
- Unemployment_Bin
- FuelPrice_Bin
- StoreSize_Bin
- Holiday_Type

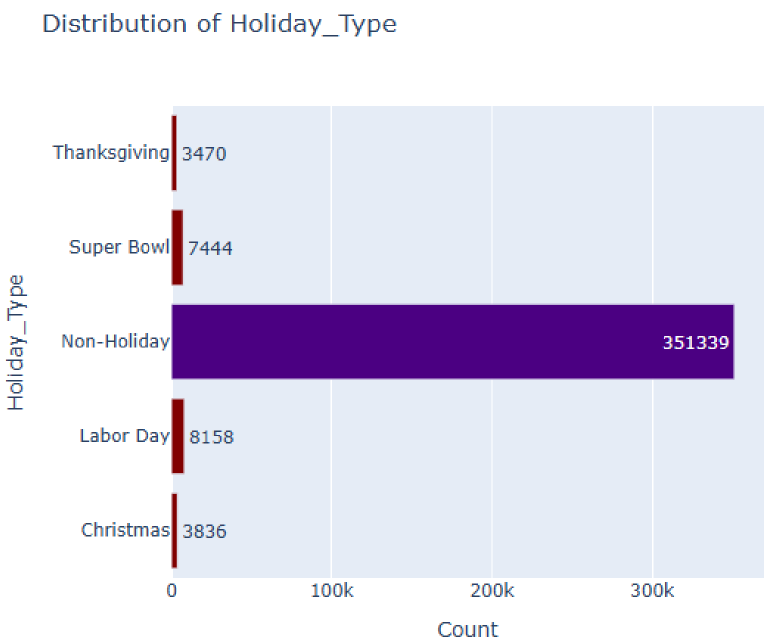
Count plots were used to visualize the frequency of each category, identify imbalances, and summarize binned features.

Type of Store:



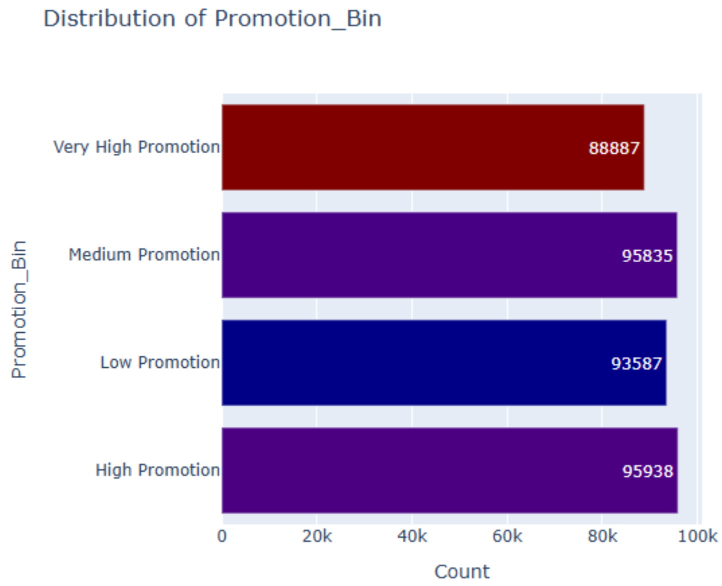
Count plot: Type A dominates with 188,222 entries, followed by Type B (149,027) and Type C (36,998). This shows an imbalance, with Type A and B stores representing most of the dataset.

Holiday Type:



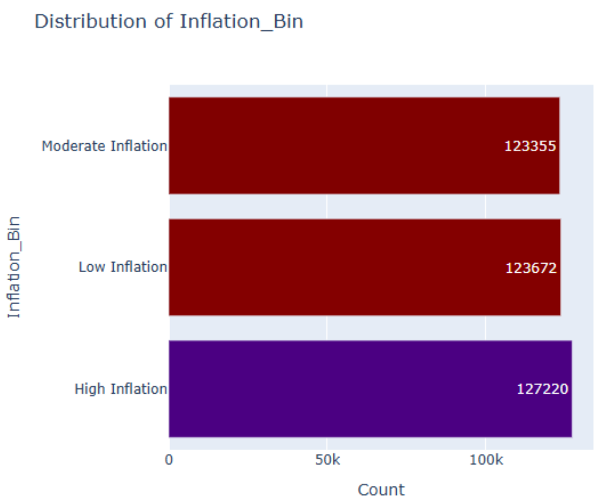
Count plot: Non-Holiday weeks account for 351,339 entries, while Labor Day (8,158), Super Bowl (7,444), Christmas (3,836), and Thanksgiving (3,470) are far less frequent. This reveals extreme class imbalance, as holidays make up only a small portion of the data.

Promotion Bin:



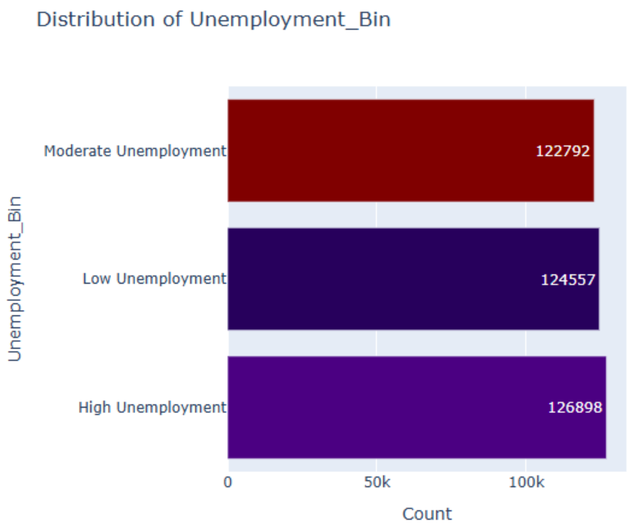
Count plot: The four categories—Low (93,587), Medium (95,835), High (95,938), and Very High (88,887) are evenly distributed, each representing roughly 23–25% of the dataset. This balanced distribution ensures unbiased analysis of promotional impact.

Inflation Bin:



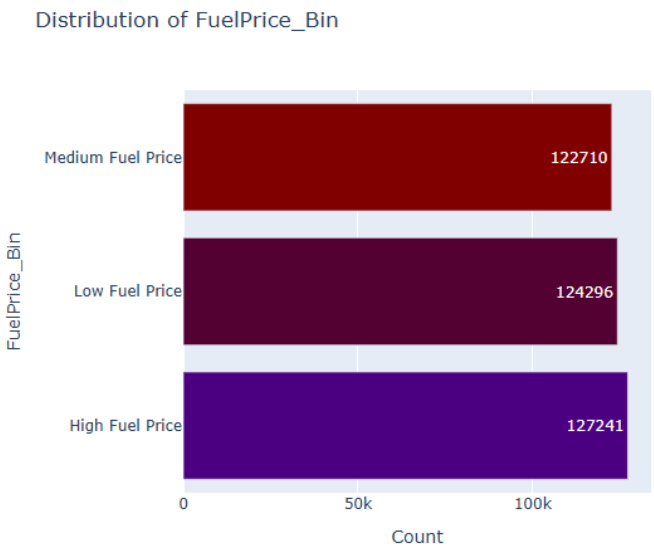
Count plot: Low (123,672), Moderate (123,355), and High (127,220) inflation categories are nearly equal, indicating no class imbalance.imbalance, as each category accounts for approximately one-third of the total dataset.

Unemployment Bin:



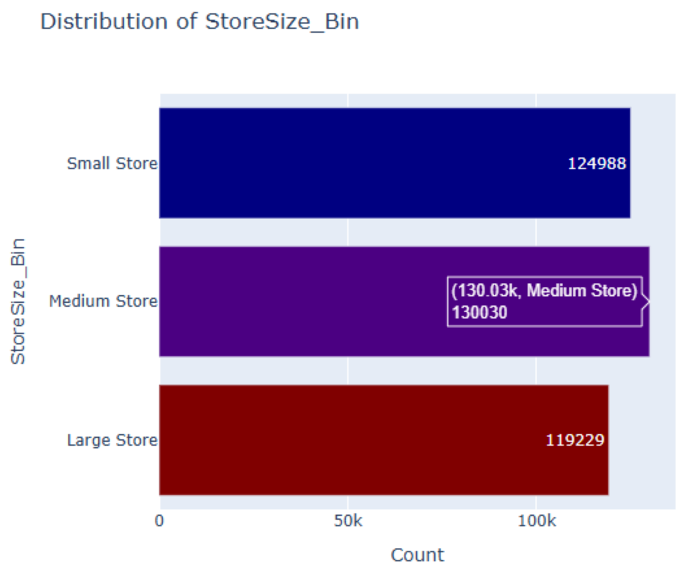
Count plot: High (126,898), Moderate (122,792), and Low (124,557) categories are well-balanced, each representing about one-third of the dataset.

Fuel Price Bin:



Count plot: High (127,241), Medium (122,710), and Low (124,296) categories are evenly distributed, preventing any price bracket from dominating the analysis.

Store Size Bin:

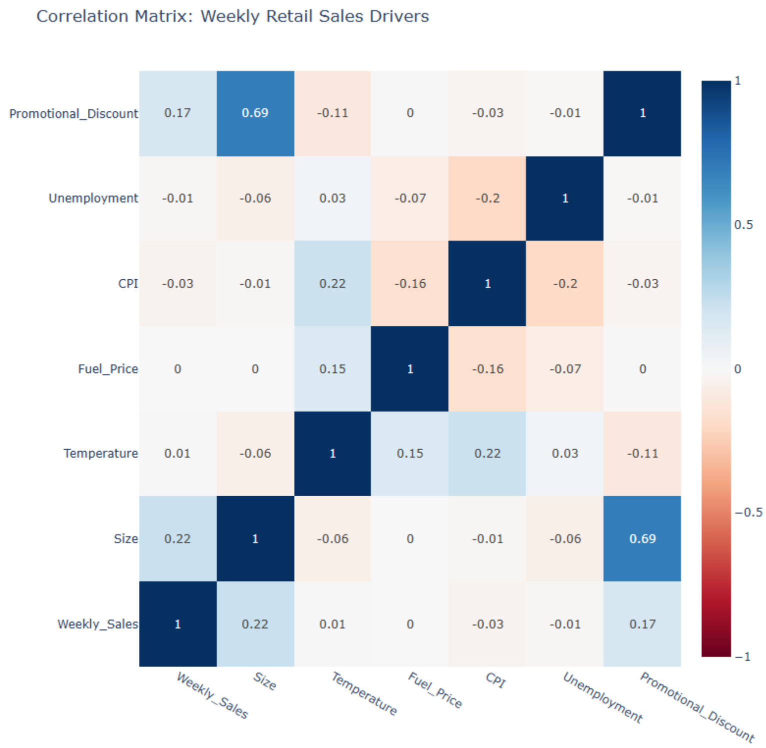


Count plot: Medium (130,030), Small (124,988), and Large (119,229) stores are well-represented, showing no significant imbalance.

6. Correlation Analysis and Kernel Density Estimation (KDE):

Before conducting detailed analysis, an initial exploration of relationships and distribution patterns among key continuous variables was carried out using correlation analysis and Kernel Density Estimation (KDE) plots. This step helps understand linear associations between variables and assess their distributions, skewness, and variability, which is important for both interpretation and future modeling.

6.1 Correlation Analysis:



Key Drivers of Weekly Sales

- **Store Size (0.22):** Store size shows the strongest positive correlation with weekly sales. Larger stores tend to generate higher revenue, likely due to greater inventory capacity and higher foot traffic.
- **Promotional Discount (0.17):** Discounts and promotions have a modest positive effect on sales. While promotions do increase revenue, the correlation is not very strong on its own.

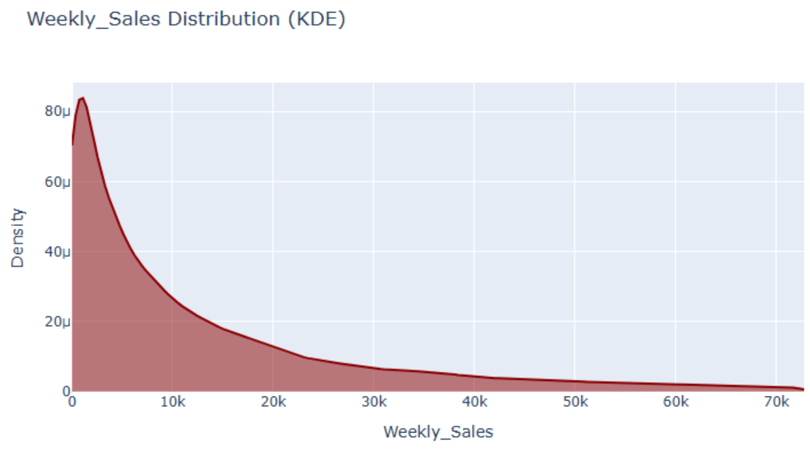
Factors with Negligible Impact
Several external economic and environmental factors show almost no linear correlation with weekly sales in this dataset:

- **Fuel Price (0.00) & Temperature (0.01):** Environmental factors have virtually no direct effect on sales.
- **CPI (-0.03) & Unemployment (-0.01):** Macroeconomic indicators like inflation and unemployment show very weak negative correlations, suggesting that small changes in the broader economy do not immediately affect weekly sales for this retailer.

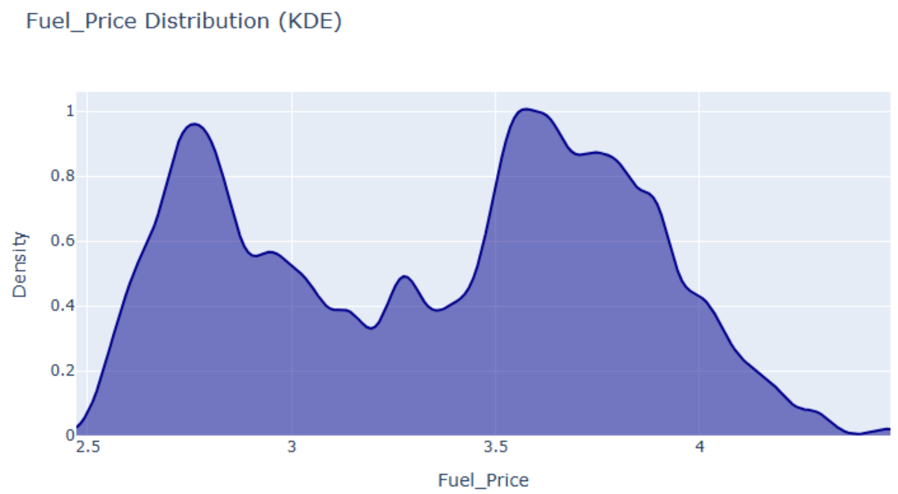
Significant Feature Interaction (Multicollinearity)

- **Store Size vs. Promotional Discount (0.69):** The strongest correlation in the matrix is between store size and promotional discount. Larger stores are more likely to offer higher discounts or have more discounted inventory. This relationship indicates that part of the effect of promotions on sales might be explained by store size, and vice versa.

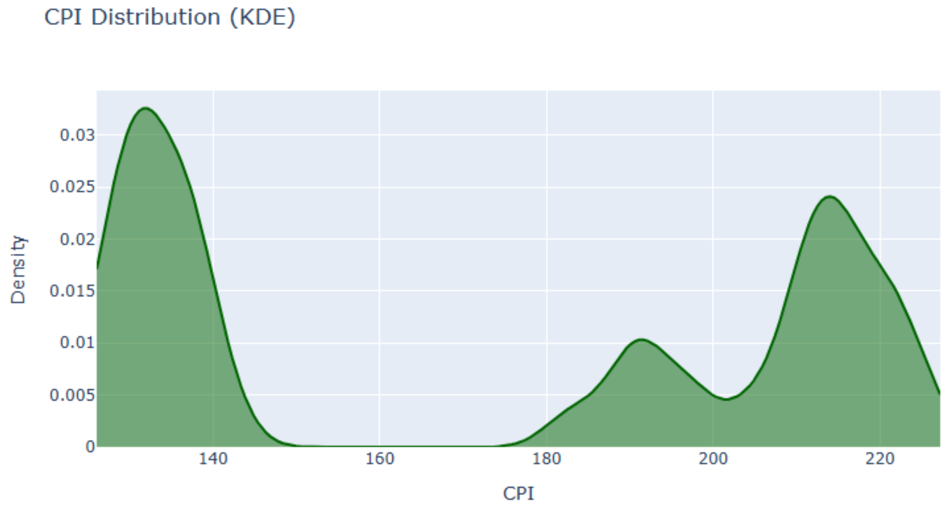
6.2 Kernel Density Estimation (KDE) Plots:



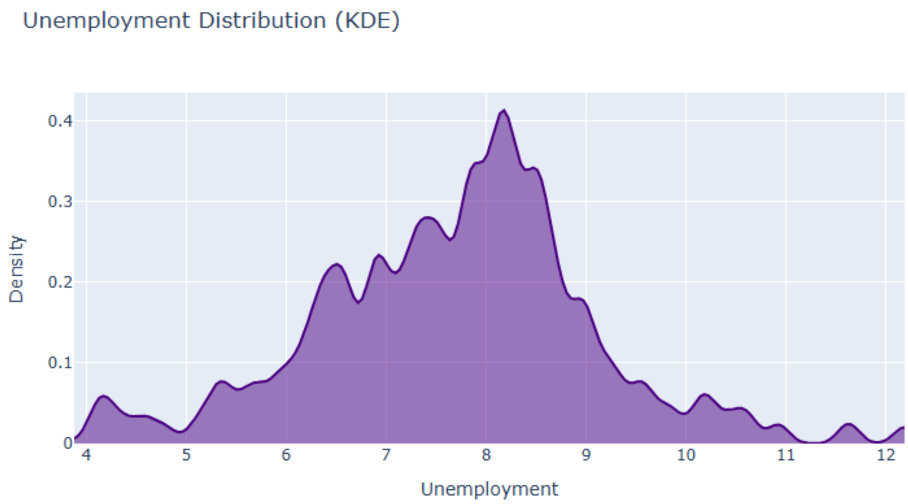
Interpretation: The KDE plot for weekly sales shows a highly right-skewed distribution, with most values concentrated at the lower end. A long tail extends toward 70k, indicating that while high-revenue weeks occur, they are rare compared to typical weekly performance.



Interpretation: Fuel prices display a clear bimodal distribution, with peaks around 2.75 and 3.60. This suggests the dataset covers two distinct periods or economic regimes where fuel prices stabilized at different levels. Using a single average would be misleading, as it falls in the valley around 3.2, where few actual observations exist.



Interpretation: CPI shows a multimodal distribution, with clusters around 130 and 215 and a noticeable gap between 150 and 180. This indicates the data was collected from regions or periods with very different inflationary conditions, highlighting a structural divide in the dataset.



Interpretation: Unemployment rates follow a unimodal, slightly right-skewed distribution, with a primary peak around 8.2%. Most values fall between 6% and 9%, but there is a long tail extending to 12%, representing occasional periods or regions with higher economic stress. Overall, the dataset reflects a generally stable but relatively high unemployment environment.

6. Univariate Analysis

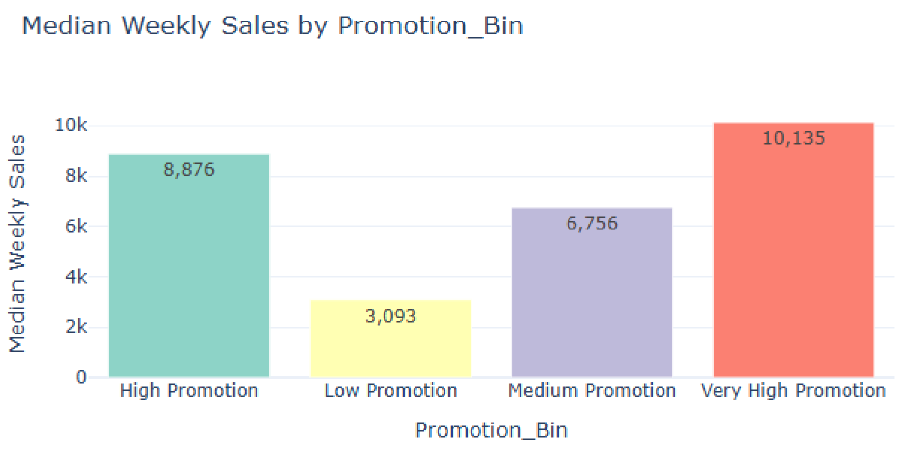
Univariate analysis examines weekly sales with respect to one explanatory factor at a time. The goal is to understand how sales vary across different promotion levels, economic conditions, store characteristics, and holiday periods without controlling for other variables. To reduce the influence of extreme values, median weekly sales were used instead of mean values.

Categorical and binned features created during feature engineering were analyzed using bar charts of median weekly sales. These visualizations provide a clear comparison of sales performance across categories.

6.1 Weekly Sales and Promotional Intensity

Weekly sales were analyzed across different levels of promotional activity using the **Promotion_Bin** variable.

Graph:

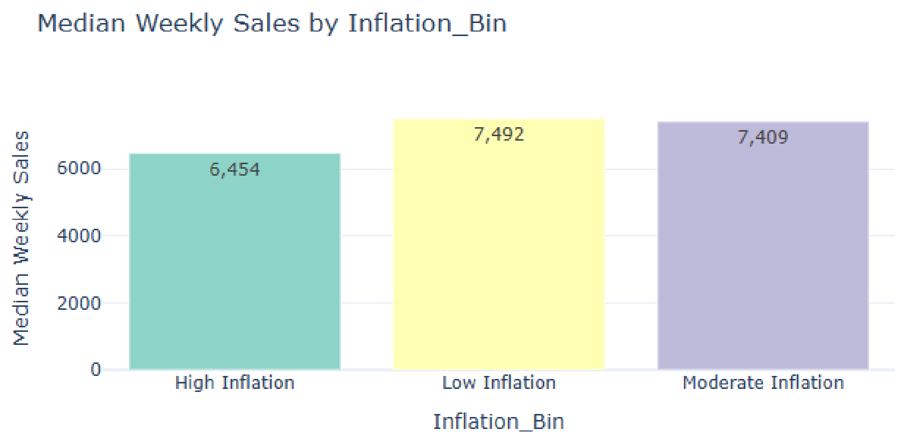


Interpretation
The bar chart shows a clear positive relationship between promotional intensity and sales. Median weekly sales increase steadily with higher promotion levels. The Very High Promotion bin reaches a peak median of 10,135, more than three times the 3,093 of the Low Promotion bin. Even a shift from Medium to High promotion results in a notable jump from 6,756 to 8,876. This indicates that promotional activities are a major driver of revenue.

6.2 Weekly Sales and Inflation Conditions

Sales were examined across **Inflation_Bin**, which categorizes CPI into low, moderate, and high inflation periods.

Graph:

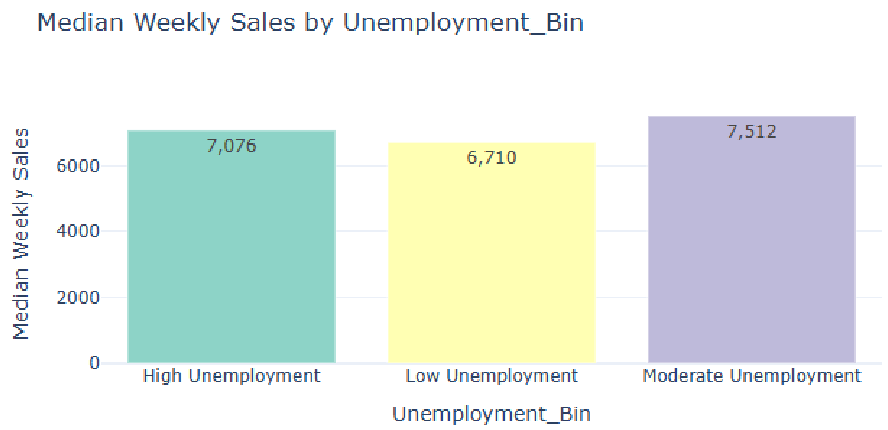


Interpretation

The bar chart suggests a slightly inverse relationship between inflation and sales. Low Inflation periods yield the highest median sales at 7,492, while Moderate Inflation remains similar at 7,409. High Inflation periods see a noticeable drop to 6,454. This implies that consumers tolerate moderate price increases, but high inflation reduces purchasing power and weekly revenue.

6.3 Weekly Sales and Unemployment Levels

The relationship between labor market conditions and sales was explored using **Unemployment_Bin**.

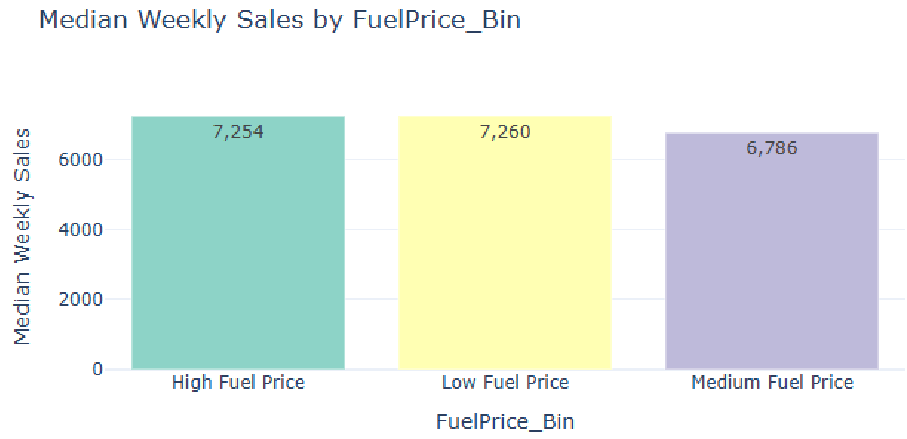


Interpretation

The chart shows a non-linear pattern. Surprisingly, Moderate Unemployment produces the highest median sales at 7,512. High Unemployment still maintains relatively strong sales at 7,076, while Low Unemployment periods show the lowest sales at 6,710. This suggests that other factors, such as store size or promotional timing, may have a stronger influence on weekly revenue than local labor market conditions.

6.4 Weekly Sales and Fuel Price Levels

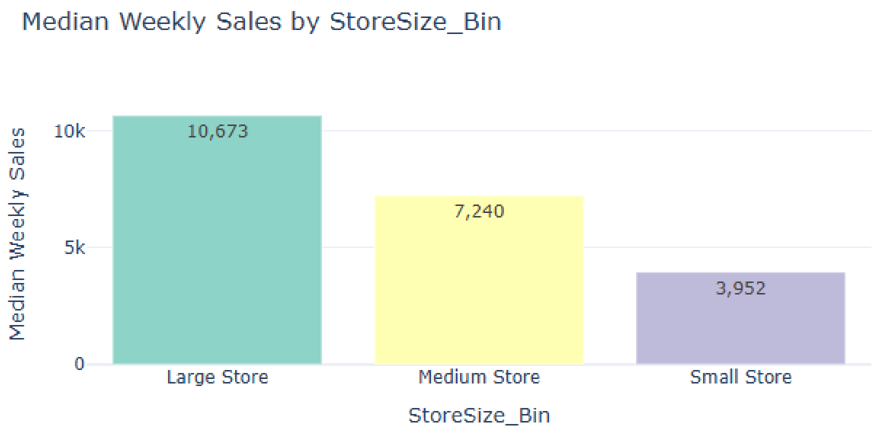
Sales were analyzed across **FuelPrice_Bin** to examine the effect of regional fuel costs on sales.



Interpretation
The bar chart shows that sales are largely stable across fuel price categories. Median sales are similar for Low Fuel Price (7,260) and High Fuel Price (7,254), while Medium Fuel Price shows a slight dip to 6,786. This indicates that customer purchasing behavior in this dataset is mostly inelastic to fuel price changes.

6.5 Weekly Sales and Store Size

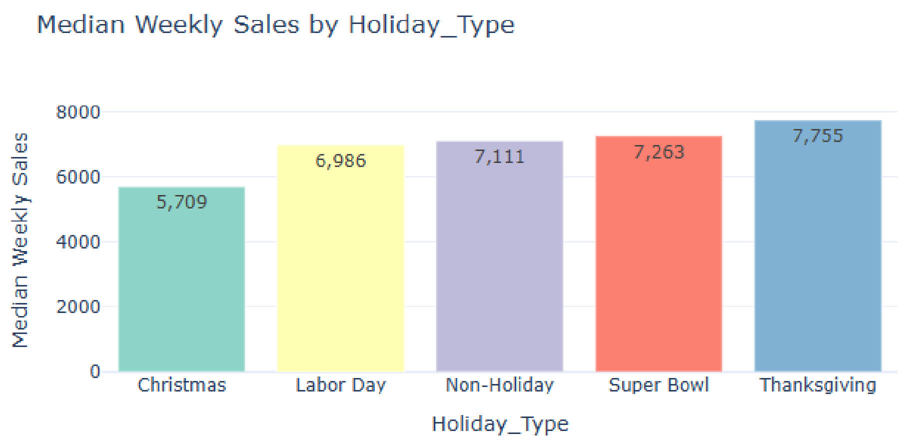
Store scale was analyzed using **StoreSize_Bin** to determine whether larger stores consistently achieve higher sales.



Interpretation
The chart shows a strong positive correlation between store size and sales. Large stores generate the highest median sales at 10,673, more than double the 3,952 of Small Stores. Medium stores fall in between at 7,240. This highlights that physical capacity and inventory breadth are major drivers of sales volume.

6.6 Weekly Sales and Holiday Effects

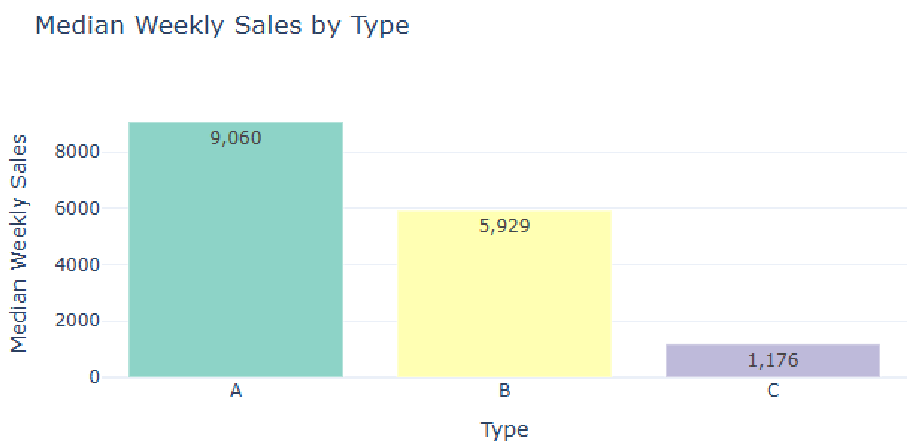
Weekly sales were examined across **Holiday_Type** to assess the impact of major U.S. holidays.



Interpretation
The bar chart shows that Thanksgiving generates the highest median sales at 7,755. Interestingly, Christmas records the lowest median sales at 5,709, even below Non-Holiday periods, which maintain a steady 7,111. This suggests that holiday effects are variable and not uniformly positive across all major holidays.

6.7 Weekly Sales and Store Type

Sales were analyzed by **Store Type** (A, B, C) to examine differences across store formats.



Interpretation
The chart shows a clear performance gap between store types. Type A stores significantly outperform the others with median sales of 9,060. Type B stores achieve a moderate 5,929, while Type C stores lag at just 1,176. This sharp decline highlights that store classification is a key determinant of revenue, reflecting differences in floor space, location, or product assortment.

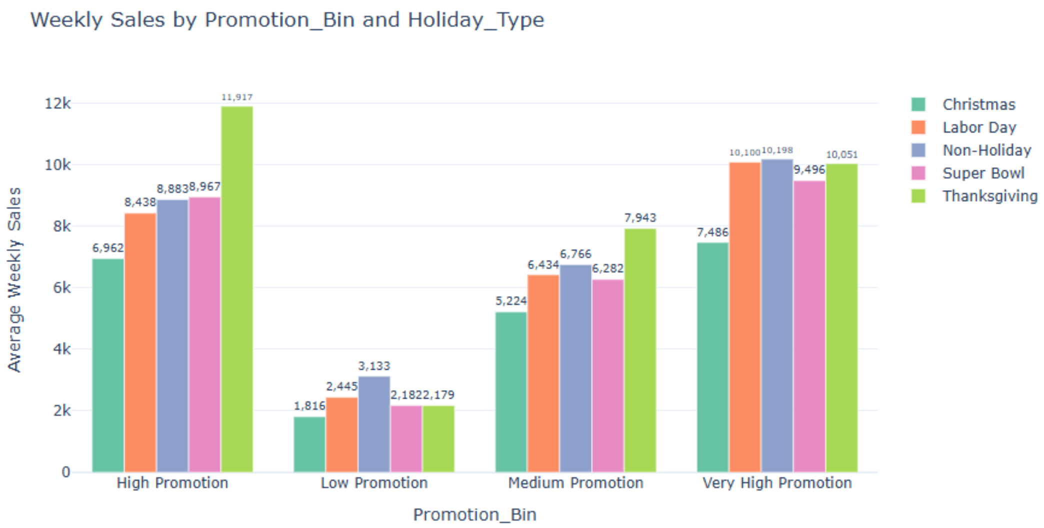
7. Bivariate Analysis

While univariate analysis highlighted the individual effects of promotions, store characteristics, and economic conditions on weekly sales, bivariate analysis examines how two factors interact simultaneously. This approach reveals whether the impact of one variable depends on the level of another, which is particularly important in retail, where promotions, store scale, seasonality, and economic factors often overlap.

Median weekly sales were used in all bivariate analyses to reduce the influence of extreme values. Grouped bar charts were employed to illustrate differences across categorical combinations clearly.

7.1 Promotional Intensity and Holiday Effects

This analysis explores the interaction between promotional intensity and holiday periods using Promotion_Bin and Holiday_Type. It assesses whether promotions are more effective during major holidays or whether holidays alone drive strong sales.



Key Patterns and Interpretation

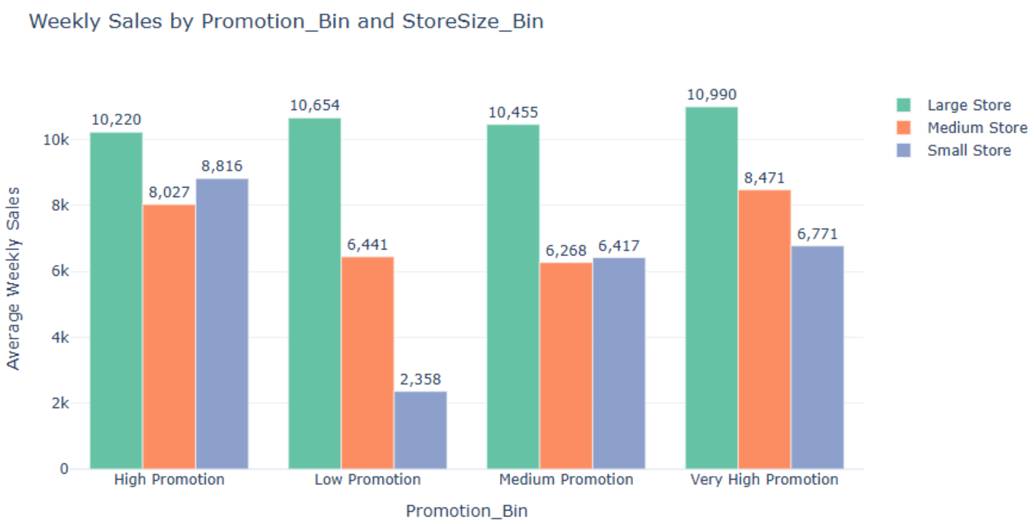
- **Promotional Dominance:** Across all holiday types, the level of promotion is the main driver of sales. Moving from Low to Very High Promotion results in a major increase, with median sales exceeding 10,000 for almost every holiday category.
- **Thanksgiving Surge:** Thanksgiving stands out when paired with aggressive promotions, reaching the highest sales of 11,917 under High Promotion.
- **Christmas Gap:** Christmas consistently underperforms within the same promotion bins. Even Very High Promotion results in only 7,486, below the Non-Holiday peak of 10,198. This may reflect store closures or logistical constraints during Christmas week.

- **Non-Holiday Resilience:** Non-Holiday weeks respond strongly to Very High Promotion (10,198), suggesting that aggressive discounts can generate “holiday-level” traffic even outside peak periods.

Strategic Insight
Promotions work universally, but their ROI is highest during Thanksgiving. Christmas promotions are less effective, so strategic planning should focus on peak-impact holidays.

7.2 Promotional Intensity and Store Size

Weekly sales were analyzed across **Promotion_Bin** and **StoreSize_Bin** to evaluate whether promotion effectiveness varies by store size.



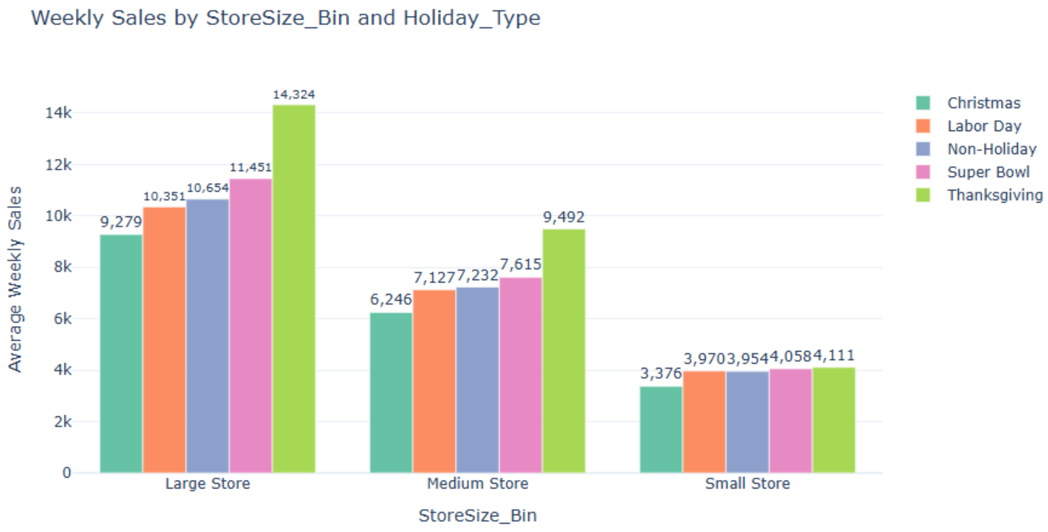
Key Patterns and Interpretation

- **Large Store Resilience:** Large Stores maintain high, stable sales regardless of promotion, ranging from 10,220 (High Promotion) to 10,990 (Very High Promotion).
- **Small Store Sensitivity:** Small Stores are highly dependent on promotions. Sales increase from 2,358 under Low Promotion to 8,816 under High Promotion.
- **High Promotion Sweet Spot:** For Small Stores, High Promotion (8,816) outperforms Very High Promotion (6,771), indicating diminishing returns at extreme discount levels.
- **Medium Store Growth:** Medium Stores show steady gains as promotions increase, from 6,441 to 8,471.

Strategic Insight
Large Stores are stable earners that require less discounting, while Small Stores need targeted, high-intensity promotions to remain competitive.

7.3 Store Size and Holiday Effects

This section examines how holiday demand varies across store sizes using **StoreSize_Bin** and **Holiday_Type**.



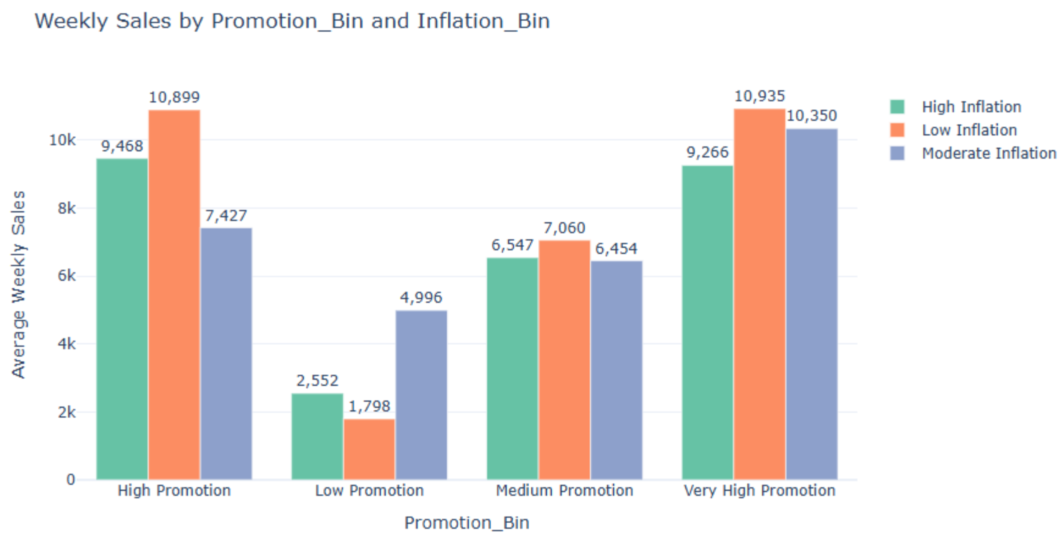
Key Patterns and Interpretation

- **Large Store Dominance:** Large Stores outperform all others, maintaining a revenue floor of at least 9,279 even during Christmas.
- **Thanksgiving Multiplier:** Thanksgiving boosts sales for all stores, but Large Stores benefit most, reaching 14,324, compared to 9,492 for Medium and 4,111 for Small Stores.
- **Consistent Holiday Trends:** Across sizes, Thanksgiving is always the top performer, followed by Super Bowl and Non-Holiday weeks, with Christmas lowest.
- **Small Store Stagnation:** Small Stores see minimal holiday-driven variation, with sales between 3,376 and 4,111, suggesting limited capacity to capitalize on seasonal traffic.

Strategic Insight
Large Stores can maximize holiday spikes, particularly during Thanksgiving. Small Stores benefit more from consistent, non-holiday promotions rather than seasonal-specific campaigns.

7.4 Promotional Intensity and Inflation Conditions

Weekly sales were analyzed across **Promotion_Bin** and **Inflation_Bin** to assess whether promotions can offset the negative effects of inflation.



Key Patterns and Interpretation

- **Low Inflation Synergy:** Highest sales occur during Low Inflation combined with High or Very High Promotion (10,899–10,935), showing consumers respond strongly to discounts when purchasing power is high.
- **Very High Promotion Equalizer:** High Promotions narrow the performance gap across inflation levels. Even under High Inflation, aggressive discounts raise sales to 9,266, approaching Moderate Inflation levels.
- **Vulnerability at Low Promotion:** Low Promotion periods see significantly lower sales. Interestingly, Moderate Inflation (4,996) outperforms Low and High Inflation in this tier.
- **High Inflation Resistance:** High Inflation suppresses sales, even under High Promotion (9,468), indicating that economic pressure limits the full effect of discounts.

Strategic Insight

Promotional effectiveness diminishes as inflation rises. Achieving similar sales during high-inflation periods requires moving from High to Very High Promotion, likely impacting profit margins.

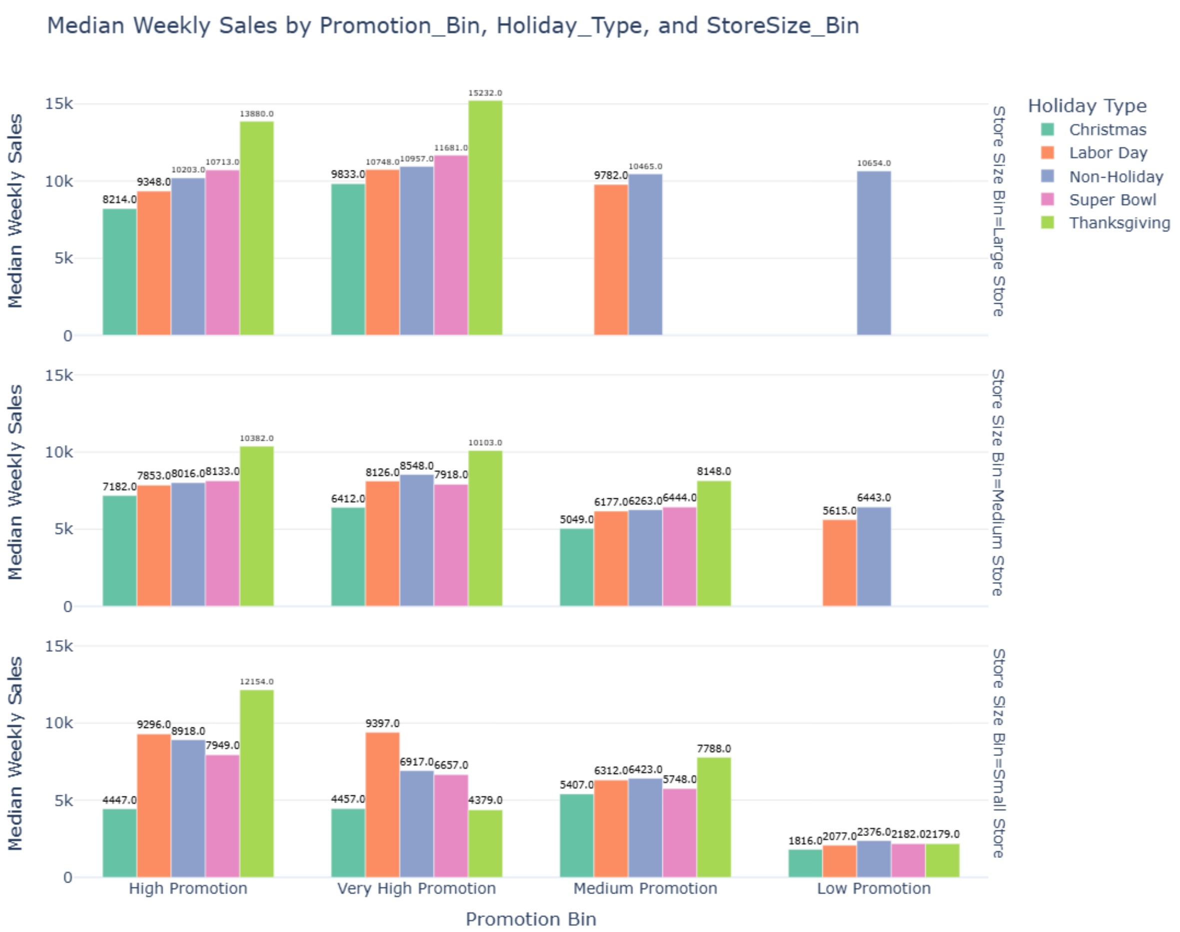
8. Multivariate Analysis

While bivariate analysis revealed important pairwise interactions affecting weekly sales, multivariate analysis examines how three factors jointly influence performance. This step is crucial because retail outcomes are rarely driven by a single factor. Instead, promotional effectiveness often depends on store characteristics and seasonal demand, particularly during major holidays.

Median weekly sales were used across all analyses to reduce the influence of extreme values and allow meaningful comparisons.

8.1 Promotional Intensity, Holiday Effects, and Store Size

This analysis explores how promotional effectiveness during holidays varies across stores of different sizes by jointly considering **Promotion_Bin**, **Holiday_Type**, and **StoreSize_Bin**. The goal is to determine whether holiday-driven gains are evenly distributed or concentrated in larger stores with greater capacity and inventory depth.



Key Patterns and Insights

- **Large Store Resilience and Peak:** Large Stores consistently dominate across all promotion tiers. Thanksgiving with Very High Promotion reaches the peak median sales of 15,232.
- **Thanksgiving as the Ultimate Multiplier:** Across all sizes, Thanksgiving paired with High or Very High promotions drives the highest revenue. Even Small Stores achieve 12,154 under High Promotion, outperforming Large Stores during Christmas in the same tier.
- **Low Promotion Floor:** Low promotions drastically reduce sales, especially for Small Stores, which drop to 1,816 during Christmas. Large Stores maintain a five-figure median (10,654) only during Non-Holiday periods.
- **Diminishing Returns on Very High Promotion:** For Small Stores, High Promotion often outperforms Very High Promotion. During Thanksgiving, sales drop from 12,154 (High) to 4,379 (Very High), suggesting extreme discounts may cannibalize revenue or reflect inventory limits.
- **Christmas Performance Paradox:** Across all tiers, Christmas is consistently the weakest holiday, often below Labor Day and Non-Holiday baselines.

Strategic Recommendation
Prioritize High Promotion for Large and Medium Stores during Thanksgiving. This combination achieves the highest revenue while potentially preserving margins compared to Very High Promotion.

8.2 Promotional Intensity, Holiday Effects, and Store Type

This analysis examines **Promotion_Bin**, **Holiday_Type**, and **Store Type (A, B, C)** to explore structural differences in store responses to promotions during holidays.

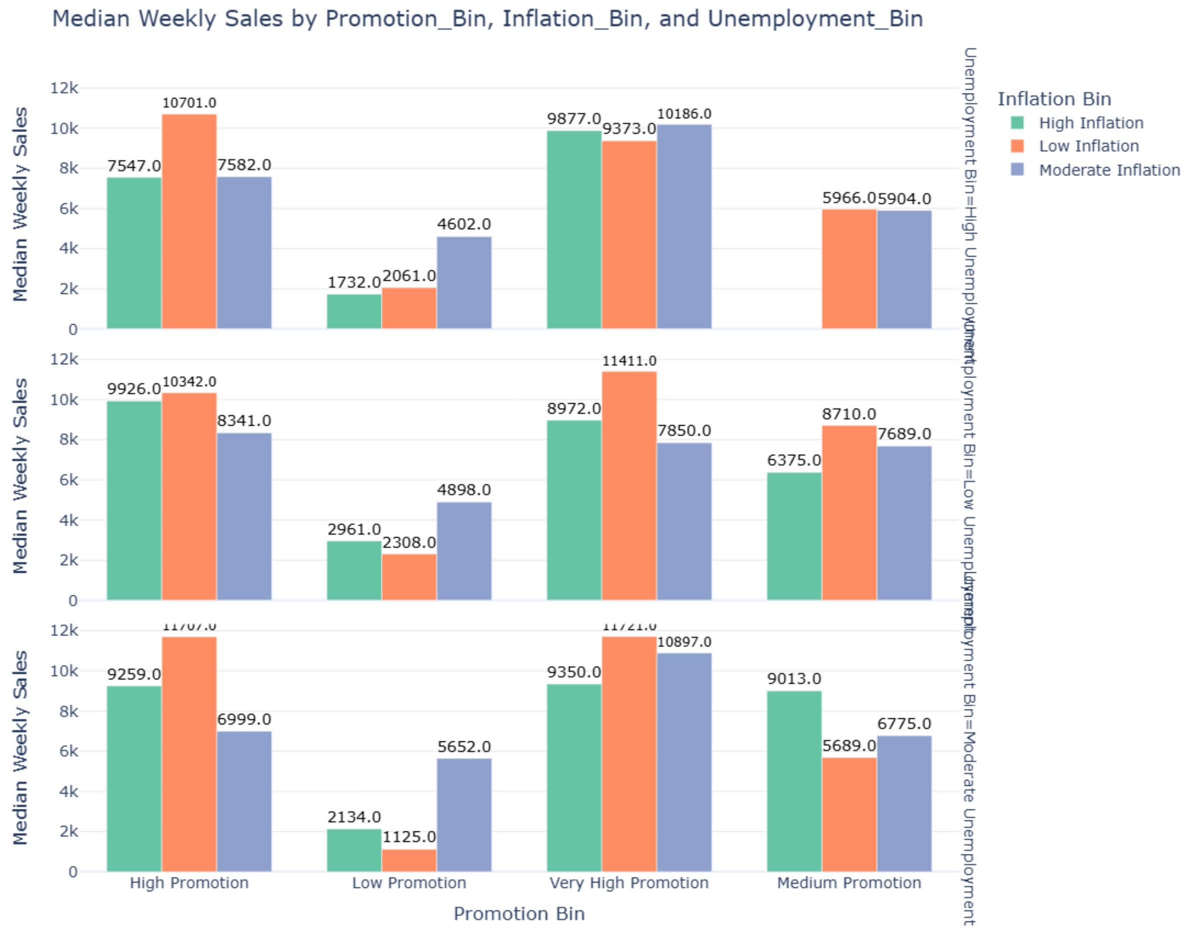
Key Patterns and Insights

- **Type A – The Strategic Multiplier:** Type A stores are the primary revenue drivers, peaking at 14,088 during Thanksgiving with Very High Promotion. The jump from High (13,797) to Very High is marginal, suggesting High Promotion can maintain volume while protecting margins.
- **Type B – Promotion-Dependent Growth:** Type B stores require aggressive discounts to achieve spikes. For instance, Labor Day sales rise from 5,809 to 8,532 when moving from Medium to Very High Promotion.
- **Type C – The Essentials Anomaly:** Type C stores perform consistently flat, with median sales between 1,130 and 1,455, regardless of holiday or promotion. These stores likely focus on everyday essentials rather than seasonal discretionary items.
- **Universal Christmas Slump:** Across Type A and B, Christmas consistently underperforms, with Type A sales 40% lower than Thanksgiving even under High Promotion.

- **Thanksgiving Dominance:** Thanksgiving remains the strongest driver, particularly for Type A stores, which capture the bulk of seasonal traffic.

8.3 Promotional Intensity, Inflation Conditions, and Unemployment Levels

This analysis investigates **Promotion_Bin**, **Inflation_Bin**, and **Unemployment_Bin** to see how macroeconomic conditions interact with promotional strategies.



Key Patterns and Insights

- **Low Inflation as a Performance Catalyst:** Low Inflation consistently boosts performance. Paired with High or Very High Promotion, median sales reach 11,707–11,721 across unemployment levels.
- **Unemployment Sweet Spot:** Moderate Unemployment environments show the highest overall sales. During Low Inflation, sales peak at 11,721, suggesting a stable middle-ground labor market supports predictable consumer behavior.
- **High Inflation Resistance:** High inflation limits promotional effectiveness. Even Very High Promotions rarely exceed 10,000 in high-inflation periods.

- **Vulnerability at Low Promotion/Low Inflation:** Surprisingly, Low Promotion under Low Inflation can result in the lowest sales (1,125), implying that consumers may require at least Medium promotions to respond.
- **Promotional Diminishing Returns:** In High Unemployment areas, increasing from High to Very High Promotion can reduce sales (10,701 → 9,373), indicating limits on consumer discretionary spending.

Strategic Takeaway
The optimal "Goldilocks" zone is Low Inflation + Moderate Unemployment + Very High Promotion. Avoid over-spending on promotions in High Unemployment areas, as revenue ceilings are naturally lower.

8.4 Fuel Price Levels, Holiday Effects, and Store Type

This analysis considers **FuelPrice_Bin**, **Holiday_Type**, and **Store Type** to explore how regional cost conditions interact with seasonality and store formats.

Key Patterns and Insights

- **Fuel Price as a Demand Stabilizer:** Type A stores remain resilient to high fuel costs, maintaining median sales above 8,000 for most holidays.
- **Christmas/High Fuel Price Collapse:** Type A stores experience a sharp sales drop during Christmas when fuel prices are High, falling to 829 median sales. This suggests logistical and transportation constraints amplify holiday vulnerabilities.
- **Thanksgiving Peak:** Thanksgiving drives the highest revenue across all fuel categories. Type A peaks at 13,285 under Low Fuel Price, highlighting the importance of transportation cost in driving bulk-shopping behavior.
- **Type C – Inelastic Essentials:** Type C sales remain flat (1,000–1,933) regardless of fuel price or holiday type, reflecting consistent demand for everyday essentials.

- **Type B – Sensitive to Fuel Costs:** Type B stores perform best under Low Fuel Price during Thanksgiving (7,827) and soften as fuel costs rise, indicating higher transportation sensitivity.

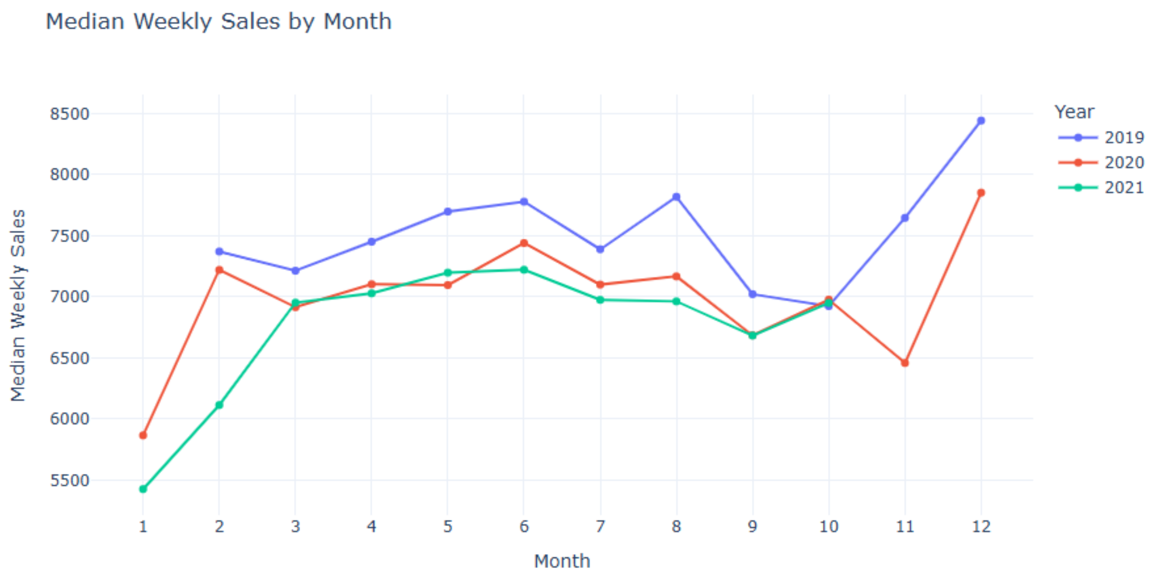
Strategic Takeaway
Type A stores are generally resilient to fuel price changes, except during extreme conditions like High Fuel Price Christmas. Type C stores provide a stable baseline, unaffected by fuel price fluctuations, while Type B stores are most sensitive to transportation costs.

9. Time and Seasonality Analysis

Given the panel and time-series nature of the dataset, a dedicated analysis of time and seasonality was conducted to identify long-term trends and recurring patterns in weekly sales. Understanding these temporal dynamics is essential for interpreting fluctuations and planning effective promotions.

9.1 Monthly Sales Trends Across Years

To examine seasonal patterns while accounting for year-to-year variation, median weekly sales were aggregated by month and year. A line plot was constructed with months on the x-axis and median weekly sales on the y-axis, with separate lines representing each year. This visualization allows direct comparison of monthly sales behavior across years and helps identify recurring peaks, troughs, or deviations driven by promotions, macroeconomic conditions, or extraordinary events.



Key Patterns and Insights

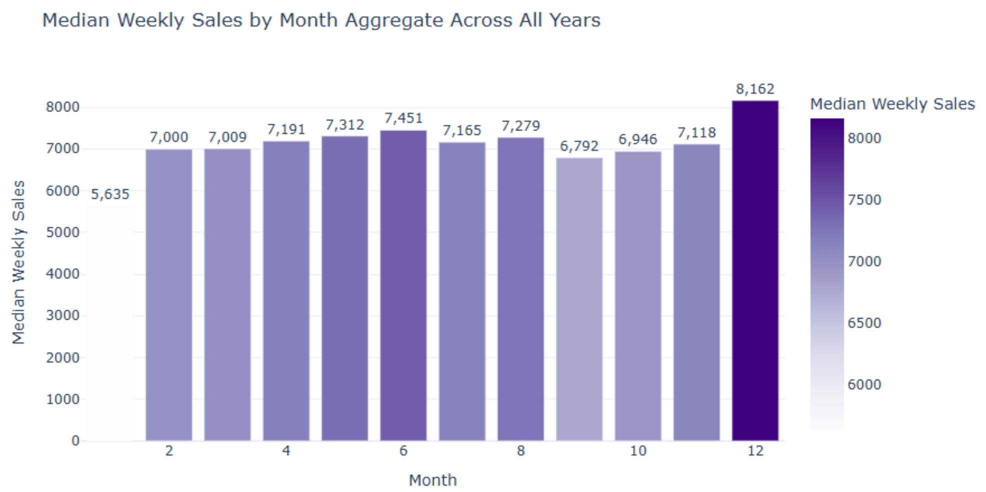
- **Year-End Peak Performance:** December consistently delivers the highest median weekly sales across all years. In 2019, the peak reaches approximately 8,450, far exceeding other months.
- **Post-Holiday Slump:** January marks the annual low, with a sharp decline in sales. Recovery occurs steadily through February and March, likely due to post-holiday restocking and early spring shopping.
- **Mid-Year Stability and Summer Bump:** Sales remain stable during spring, with a noticeable secondary peak in August, likely driven by "Back to School" or summer travel-related demand.
- **November Divergence:** November shows a dip just before December, suggesting that consumers may hold back spending in anticipation of major holiday discounts.

- **Overall Trend Decline:** While seasonal shapes are consistent, baseline sales decline over the three-year period. 2019 consistently sits at the top, while 2021 tracks lower, indicating a general cooling of weekly revenue.

Strategic Takeaway
The business is highly dependent on the Q4 holiday window to meet annual targets. The consistent August peak suggests a reliable opportunity for a "pre-holiday" promotional push to bridge the summer-to-winter gap.

9.2 Aggregate Monthly Seasonality

To isolate overall seasonality independent of year-specific effects, median weekly sales were aggregated by month across all years. A bar chart summarizes typical sales performance for each month, highlighting consistent seasonal patterns useful for inventory planning and promotional timing.



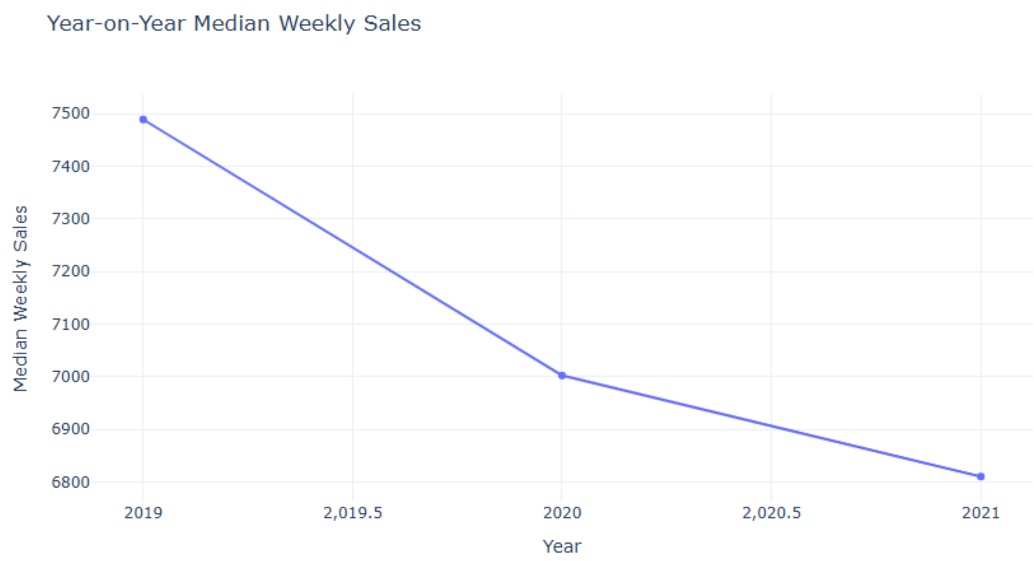
Key Patterns and Insights

- **December Peak:** December remains the definitive peak, with median weekly sales of 8,162—the only month above 8,000—representing a massive seasonal surge.
- **January Floor:** Following the holiday season, January records the lowest median sales at 5,635, reflecting post-holiday spending fatigue.
- **Summer Plateau:** April through August shows stable sales between 7,165 and 7,451, peaking in June at 7,451, indicating steady mid-year demand.
- **Gradual Q3/Q4 Build-Up:** After a slight dip in September (6,792), sales rise gradually through October (6,946) and November (7,118), culminating in the December surge. November includes Thanksgiving, but the largest spike occurs in the final weeks of the year.

Strategic Takeaway
The business exhibits a "U-shaped" annual recovery. The mid-year plateau provides a reliable revenue baseline, but strategies must focus on mitigating the January slump and maximizing the December peak.

9.3 Year-on-Year Sales Trends

To assess long-term trends, median weekly sales were aggregated at the yearly level and visualized using a line plot. This approach abstracts from seasonal fluctuations to capture overall growth or decline.



Key Patterns and Insights

- **Consistent Annual Decline:** Median weekly sales decline steadily over three years from 7,490 in 2019 to 7,000 in 2020, and further to 6,810 in 2021.
- **Deceleration of the Drop:** The rate of decline slows slightly over time, with a 490-unit drop between 2019 and 2020 and a smaller 190-unit decline between 2020 and 2021.
- **Long-Term Revenue Pressure:** Seasonal peaks, including December, are insufficient to offset the overall downward trend.

Strategic Takeaway
The overall sales "pie" is shrinking. Combined with earlier findings that High Inflation and High Fuel Prices suppress sales, it is evident the business faces a challenging economic environment. Optimizing high-margin Type A stores during peak periods, particularly Thanksgiving, will be critical to compensate for the declining annual baseline.